

Database Foundations

5-2: Mapping Primary and Foreign Keys

Practice

Exercise 1: Observe the mapping of the unique identifiers and relationship in the Relational Model

Overview

In this practice you will observe the mapping of the unique identifiers and relationship in the Academic Database Relational Model.

Tasks

1. Compare the Logical Model and the engineered Relational Model to verify:
 - a. The Unique Identifiers that have been mapped as Primary Keys
 - b. The Unique Identifiers that have been mapped as Unique Keys
 - c. The Relationships that have been mapped as Foreign Keys

Exercise 2: Define table name abbreviations in csv file

Overview

In this practice, you define abbreviations for keys and constraints in a .csv file.

Tasks

1. To define the abbreviations for table names, perform the following steps:
 - a. Open a spreadsheet application
 - b. In the first column list plural table names, and in the second column the required abbreviation for each table.
 - c. Save the file as .csv and note the location.

Exercise 3: Define Name Template

Overview

In this practice, you define templates (name patterns) for keys, and constraints by using combinations of predefined variables.

Tasks

1. You can set a template for the keys, indexes, and constraints in table or entity by using combinations of predefined variables. To define the name patterns perform the following steps:
 - a. Right-click the Academic Database design in the Object Browser and select **Properties**. Expand **Settings > Naming Standard** and select **Templates**.
 - b. Set the predefined variables as follows:

Table Constraints

Element	Pre-Defined Variable
Primary Key	{table abbr}_PK
Foreign Key	{child abbr}_{parent abbr}_FK
Check Constraint	{table}_CK
Unique Constraint	{table abbr}_{column}_UK
Index	{table}_{column}
Automatic Index	{table}_{column}
Column Check Constraint	{table}_{column}
Not Null Constraint	{table abbr}_{column}_NN
Column Foreign Key	{ref table abbr}_{ref column}
Surrogate Key	{table abbr}_PK
Surrogate Key Column	{table abbr}_ID
Discriminator Column	{table abbr}_TYPE

Exercise 4: Apply Name Template to the Relational Model

Overview

After defining the Naming Template you can apply it to one entity/table or the entire Logical Model/Relational Model. In this practice, you will apply the Naming Template to the entire Relational Model.

Tasks

1. To apply the template to the entire Relational model, perform the following:
 - a. Click **Tools > Name abbreviations**.
 - b. Browse to the .csv file containing the abbreviations.
 - c. Un-check Tables (to maintain existing names from the Glossary), and then click OK.

Note: if the Logical model has already been engineered, sometimes it may be necessary to delete all objects in the relational model (open the relational tab, click Edit menu option > Select All, then press the delete key), re-engineer the logical, then apply the naming abbreviations.

Exercise 5: Select how subtypes are generated in the Relational Model

Overview

In this practice, you will define how subtypes are mapped to the Academic Database Relational Model.

Tasks

1. To define how subtypes are mapped to the Academic Database Relational Model, perform the following steps:
 - a. Click the Logical tab.
 - b. Double click the Faculty Super type entity to edit properties
 - c. Select Subtypes from the Options in the Left pane.
 - d. From the Subtree Generation drop down option, select Single Table. Click OK.
 - e. Re-engineer to Relational Model.